# PHISHING EMAIL ANALYSIS REPORT

For Future Interns

PRESENTED BY :

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# 1. Executive Summary

This report presents the findings of a phishing email analysis conducted as part of Cyber Security Task 2 (2026) for Future Interns. Three email samples were collected and subjected to thorough analysis including email header examination, sender domain verification, link inspection, and VirusTotal URL scanning.

Phishing remains the most prevalent cyber attack vector against organizations globally. Attackers exploit human psychology — urgency, fear, and trust — to bypass even robust technical defenses. This report equips employees and security teams with the knowledge to detect, classify, and report phishing attempts effectively.

# 2. Scope & Objectives

## 2.1 Scope

Three email samples were analyzed for this report:

- Email 1 – Fake System Update Notification
- Email 2 – Legitimate Google Security Notification
- Email 3 – Fake Account Verification Request

## 2.2 Objectives

- Analyze email headers, sender domains, and embedded links
- Identify phishing indicators in each email sample
- Classify each email as Safe, Suspicious, or Phishing
- Provide user-friendly explanations of each threat
- Deliver prevention guidelines and awareness tips

# 3. Methodology

Each email was analyzed using the following approach:

- Sender Domain Inspection — Verification of sender email domain against known legitimate domains
- Email Header Analysis — Using Google Admin Toolbox to inspect SPF, DKIM, and DMARC authentication results
- Link Analysis — Inspection of URLs using VirusTotal for malicious content detection
- Tone & Language Analysis — Identification of urgency, fear-based language, and impersonation cues
- Content Review — Assessment of greeting personalization, formatting, and legitimacy

# 4. Email Analysis & Findings

## 4.1 Email Sample 1 — Fake System Update Notification

| Field | Details |
|---|---|
| **From** | update@download-secure-update.com |
| **To** | exampleuser@gmail.com |
| **Subject** | Critical Update Required – Install Immediately |
| **Link** | http://download-secure-update.com/install-update.exe |
| **Risk** | **PHISHING** |

### Email Header Analysis — Email 1:

| | |
|---|---|
| **MessageId** | 9988776655@download-secure-update.com |
| **Created at:** | 4/21/2026, 4:45:52 PM GMT+5:30 ( Delivered after 3 sec ) |
| **From:** | "System Update" <update@download-secure-update.com> |
| **To:** | exampleuser@gmail.com |
| **Subject:** | Critical Update Required – Install Immediately |
| **SPF:** | fail with IP Unknown! Learn more |
| **DKIM:** | fail with domain download-secure-update.com Learn more |
| **DMARC:** | fail Learn more |

| # | Delay | From * | | | To * | Protocol | Time received |
|---|---|---|---|---|---|---|---|
| 0 | 1 sec | unknown | → | | download-secure-update.com | ESMTP | 4/21/2026, 4:45:53 PM GMT+5:30 |
| 1 | 2 sec | download-secure-update.com. | → | Google | mx.google.com | ESMTPS | 4/21/2026, 4:45:55 PM GMT+5:30 |
| 2 | | | → | Google | 2002:a05:6a10:777:b0:7c8:yzab:3344 | SMTP | 4/21/2026, 4:45:55 PM GMT+5:30 |

**Key Header Findings:**

- SPF: FAIL — IP Unknown. The sending IP is not authorized to send on behalf of the domain
- DKIM: FAIL — Signature verification failed for domain download-secure-update.com
- DMARC: FAIL — Email fails both SPF and DKIM, marking it as unauthenticated
- Origin: Sent from an unknown server routed through Google's mail infrastructure

**Phishing Indicators:**

- Suspicious sender domain — not a recognized OS or software vendor
- Malicious .exe file delivered over plain HTTP (no encryption)
- Urgency language — 'Critical', 'Install Immediately', '24 hours'
- Generic greeting — 'Dear User' (no personalization)
- Threatens 'limited system functionality' to coerce action

**Attack Explanation:**

This email tricks the user into downloading and running a malicious executable file disguised as a system update. Once installed, the malware could give attackers remote access, steal passwords, or lock files for ransom. The attacker uses fear of data loss and a fake deadline to prevent the user from thinking critically before clicking.

## 4.2 Email Sample 2 — Google Security Notification

| Field | Details |
|---|---|
| **From** | no-reply@accounts.google.com |
| **To** | exampleuser@gmail.com |
| **Subject** | Security Notification: Password Changed Successfully |
| **Link** | https://accounts.google.com/security |
| **Risk** | **SAFE / LEGITIMATE** |

**Email Header Analysis — Email 2:**

| | |
|---|---|
| **MessageId** | xyz987abc654@mail-pj1-f45.google.com |
| **Created at:** | 4/22/2026, 8:25:27 PM GMT+5:30 ( Delivered after 3 sec ) |
| **From:** | "Google Account" <no-reply@accounts.google.com> |
| **To:** | exampleuser@gmail.com |
| **Subject:** | Security Notification: Password Changed Successfully |
| **SPF:** | **pass** with IP 209.85.216.45<br>Learn more |
| **DKIM:** | **pass** with domain accounts.google.com<br>Learn more |
| **DMARC:** | **pass**<br>Learn more |

| # | Delay | From * | | | To * | Protocol | Time received |
|---|---|---|---|---|---|---|---|
| 0 | 1 sec | mail-pj1-f45.google.com | → | [Google] | mail-pj1-f45.google.com | ESMTP | 4/22/2026, 8:25:28 PM GMT+5:30 |
| 1 | 2 sec | mail-pj1-f45.google.com. | → | [Google] | mx.google.com | ESMTPS | 4/22/2026, 8:25:30 PM GMT+5:30 |
| 2 | | | → | [Google] | 2002:a17:906:1111:b0:aaa:bbbb:cccc | SMTP | 4/22/2026, 8:25:30 PM GMT+5:30 |

### Key Header Findings:

- SPF: PASS — IP 209.85.216.45 is an authorized Google mail server
- DKIM: PASS — Signature verified for domain accounts.google.com
- DMARC: PASS — Email is fully authenticated as originating from Google
- Routing: Sent from mail-pj1-f45.google.com through Google infrastructure

### Legitimacy Indicators:

- Verified domain — accounts.google.com is Google's official account domain
- HTTPS link pointing to a known Google property
- No coercion — states 'no further action required' if user made the change
- Calm, informational tone — empowers the user rather than pressuring them
- No executable attachments or suspicious redirects

## 4.3 Email Sample 3 — Fake Account Verification Request

| Field | Details |
|---|---|
| **From** | support@secure-login-update.com |
| **To** | exampleuser@gmail.com |
| **Subject** | Urgent: Verify Your Account Immediately |
| **Link** | http://secure-login-update.com/verify-account |
| **Risk** | **PHISHING** |

### Email Header Analysis — Email 3:

| | |
|---|---|
| **MessageId** | 4455778899@secure-login-update.com |
| **Created at:** | 4/23/2026, 7:10:09 PM GMT+5:30 ( Delivered after 3 sec ) |
| **From:** | "Account Security" <support@secure-login-update.com> |
| **To:** | exampleuser@gmail.com |
| **Subject:** | Urgent: Verify Your Account Immediately |
| **SPF:** | fail with IP Unknown! Learn more |
| **DKIM:** | fail with domain secure-login-update.com Learn more |
| **DMARC:** | fail Learn more |

| # | Delay | From * | | | To * | Protocol | Time received |
|---|---|---|---|---|---|---|---|
| 0 | 1 sec | unknown | → | | secure-login-update.com | ESMTP | 4/23/2026, 7:10:10 PM GMT+5:30 |
| 1 | 2 sec | secure-login-update.com. | → | [Google] | mx.google.com | ESMTPS | 4/23/2026, 7:10:12 PM GMT+5:30 |
| 2 | | | → | [Google] | 2002:a05:6a10:888:b0:8d9:cdef:5566 | SMTP | 4/23/2026, 7:10:12 PM GMT+5:30 |

### Key Header Findings:

- SPF: FAIL — IP Unknown. Sender IP not authorized by domain
- DKIM: FAIL — Signature verification failed for domain secure-login-update.com
- DMARC: FAIL — Email fails all authentication checks

- Origin: Unknown server, no verifiable sending infrastructure

**Phishing Indicators:**

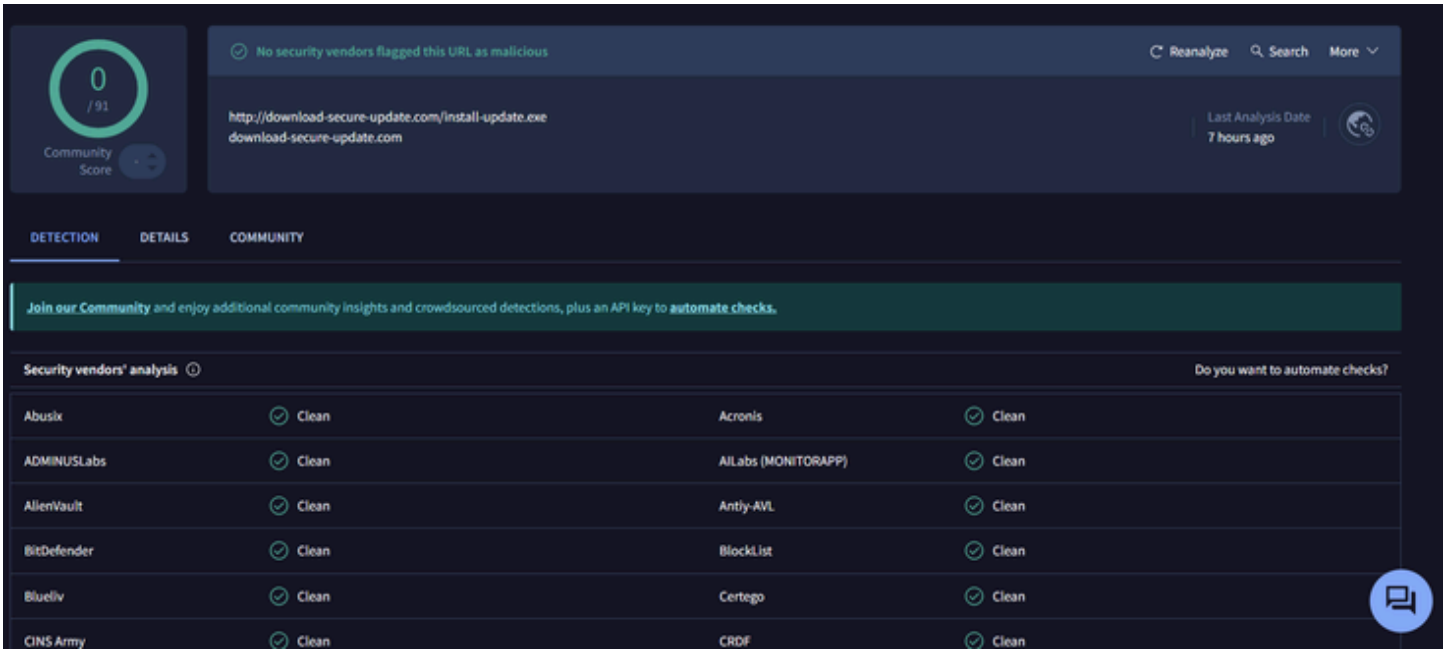
- Fraudulent domain — 'secure-login-update.com' uses trust keywords to appear legitimate
- Vague reference — 'your account' without specifying which service
- Threat of permanent suspension within 24 hours
- Verification link over HTTP — no security certificate
- Generic greeting — 'Dear User' (mass-targeted)

**Attack Explanation:**

This is a credential harvesting attack. The link leads to a fake login page that looks like a real service. When the user enters their credentials, the attacker captures them instantly and uses them to access the real account — potentially leading to identity theft, financial fraud, or attacks on the victim's contacts.

## 5. VirusTotal URL Analysis

The malicious link from Email 1 (http://download-secure-update.com/install-update.exe) was submitted to VirusTotal for multi-engine scanning. The results are presented below.



| | | | |
|---|---|---|---|
| Criminal IP | Clean | CTX AI | Clean |
| Cyble | Clean | CyRadar | Clean |
| desenmascara.me | Clean | DNS8 | Clean |
| Dr.Web | Clean | EmergingThreats | Clean |
| Emsisoft | Clean | ESET | Clean |
| ESTsecurity | Clean | Fortinet | Clean |
| G-Data | Clean | Google Safebrowsing | Clean |
| GreenSnow | Clean | Heimdal Security | Clean |
| IPsum | Clean | Juniper Networks | Clean |
| LevelBlue | Clean | Lionic | Clean |
| Malwared | Clean | MalwarePatrol | Clean |
| OpenPhish | Clean | Phishing Database | Clean |
| Phishtank | Clean | PREBYTES | Clean |
| Quick Heal | Clean | Quttera | Clean |
| Rising | Clean | Sangfor | Clean |

| | | | |
|---|---|---|---|
| Scantitan | Clean | Seclookup | Clean |
| securolytics | Clean | Sophos | Clean |
| StopForumSpam | Clean | Sucuri SiteCheck | Clean |
| ThreatHive | Clean | URLhaus | Clean |
| Viettel Threat Intelligence | Clean | ViriBack | Clean |
| VX Vault | Clean | Webroot | Clean |
| Yandex Safebrowsing | Clean | ZeroCERT | Clean |
| 0xSI_f33d | Unrated | alphaMountain.ai | Unrated |
| AlphaSOC | Unrated | ArcSight Threat Intelligence | Unrated |
| AutoShun | Unrated | Blone.Ai PreCrime | Unrated |
| Bkav | Unrated | ChainPatrol | Unrated |
| Chong Lua Dao | Unrated | Cluster25 | Unrated |
| CSIS Security Group | Unrated | Cyan | Unrated |
| Ermes | Unrated | Forcepoint ThreatSeeker | Unrated |
| GCP Abuse Intelligence | Unrated | GreyNoise | Unrated |

| | | | |
|---|---|---|---|
| CSIS Security Group | ? Unrated | Cyan | ? Unrated |
| Ermes | ? Unrated | Forcepoint ThreatSeeker | ? Unrated |
| GCP Abuse Intelligence | ? Unrated | GreyNoise | ? Unrated |
| Gridinsoft | ? Unrated | Guardpot | ? Unrated |
| Hunt.io Intelligence | ? Unrated | Kaspersky | ? Unrated |
| Lumu | ? Unrated | MalwareURL | ? Unrated |
| Mimecast | ? Unrated | Netcraft | ? Unrated |
| PhishFort | ? Unrated | PhishLabs | ? Unrated |
| PrecisionSec | ? Unrated | SafeToOpen | ? Unrated |
| Sansec eComscan | ? Unrated | Snort IP sample list | ? Unrated |
| SOCRadar | ? Unrated | URLQuery | ? Unrated |
| VIPRE | ? Unrated | Xcitium Verdict Cloud | ? Unrated |
| ZeroFox | ? Unrated | | |

### VirusTotal Analysis Interpretation:

Although VirusTotal returned 0/91 detections, this does NOT confirm the URL is safe. A clean scan result is common for newly registered phishing domains that have not yet been indexed by threat intelligence databases. This technique is known as a 'fresh domain' attack — attackers register new domains specifically to evade blocklists.

- The domain download-secure-update.com does not belong to any legitimate software vendor
- The link serves a direct .exe file over HTTP — a clear red flag regardless of scan results
- SPF, DKIM, and DMARC all failed — confirming the domain is not a legitimate sender
- Multiple vendors are listed as 'Unrated' — meaning the domain is too new for their databases
- Conclusion: Clean VirusTotal result + failed email authentication = high confidence phishing

## 6. Risk Classification Summary

| # | Sender Domain | Subject | Risk Level | Threat Type |
|---|---|---|---|---|
| 1 | download-secure-update.com | Critical Update Required | **PHISHING** | Malware Download |
| 2 | accounts.google.com | Password Changed Successfully | **SAFE** | Legitimate Notification |
| 3 | secure-login-update.com | Urgent: Verify Your Account | **PHISHING** | Credential Harvesting |

## 7. Common Phishing Techniques Observed

| Technique | Description |
|---|---|
| **Domain Spoofing** | Using lookalike domains to impersonate trusted organizations. |
| **Urgency & Fear** | Artificial deadlines and threats of suspension to pressure quick action. |
| **Malicious Links** | Embedded links to fake sites or malware — often over HTTP without encryption. |
| **Generic Greetings** | 'Dear User' instead of real name — sign of mass-targeted phishing. |
| **Auth Failures** | SPF, DKIM, and DMARC all fail — email is not from a legitimate authenticated source. |
| **Fresh Domain Attack** | New domains bypass VirusTotal and threat blocklists before being indexed. |

## 8. Prevention & Awareness Guidelines

### 8.1 What Users SHOULD Do

- Always check the full sender email domain before clicking any link

- Hover over links to preview the actual URL destination before clicking
- Look for HTTPS in links — avoid clicking HTTP links in emails
- Visit official websites directly rather than using email links
- Report suspicious emails to your IT or security team immediately
- Enable Multi-Factor Authentication (MFA) on all accounts
- Keep your OS and antivirus software up to date

### 8.2 What Users Should NOT Do

- Do NOT click download links asking you to install software via email
- Do NOT enter passwords on sites you reached by clicking an email link
- Do NOT act on '24-hour' deadlines without independently verifying
- Do NOT trust emails that don't address you by your real name
- Do NOT open .exe, .zip, or .js attachments from unknown senders
- Do NOT assume a clean VirusTotal result means a URL is safe

### 8.3 For Organizations

- Implement SPF, DKIM, and DMARC email authentication protocols
- Conduct regular phishing simulation exercises for all staff
- Deploy email filtering and anti-phishing security gateways
- Create a simple, clear process for employees to report suspicious emails
- Run periodic security awareness campaigns and refresher training

## 9. Employee Quick Reference — Do's & Don'ts

| DO | DON'T |
|---|---|
| Verify the sender's full email domain | Click links or download files without checking |
| Hover over links to check the real URL | Trust emails that create urgency or panic |
| Check SPF/DKIM/DMARC authentication results | Assume a clean VirusTotal scan means safety |
| Report suspicious emails to IT/Security | Enter credentials via links in emails |
| Use MFA on all important accounts | Open .exe or unknown file attachments |
| Go directly to official websites for account issues | Assume an email is safe based on its appearance alone |

## 10. Conclusion

This analysis demonstrates that phishing emails are designed to exploit human psychology rather than technical weaknesses. Of the three emails reviewed, two were confirmed phishing attacks and one was a legitimate Google notification.

Email header analysis proved especially valuable — both phishing emails showed complete failure of SPF, DKIM, and DMARC authentication, while the legitimate Google email passed all three checks. The VirusTotal scan also highlighted an important insight: a clean URL scan does not confirm safety, particularly for newly registered phishing domains.

A strong defense combines user awareness with technical controls — SPF/DKIM/DMARC enforcement, endpoint protection, and email filtering. Security is a shared responsibility between technology and the people who use it.